IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

Corrected Copy

POWER ENGINEERING

Question 2 a(iii) ← typo
Should be 2 a(ii)

Monday, 3 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri

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Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_O}{V_I} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fLI_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_O}{V_I} = \frac{1}{1-\delta}$ or $\frac{V_O}{V_I} = \frac{1}{1-\frac{V_I\delta^2}{2fLI_I}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\,V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\,I_P$

Power $P_{3\phi} = 3\,V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Induction Machines

$\omega_S = {}^{\omega_E}/_{P}$; $s = \frac{\omega_S - \omega_R}{\omega_S}$; $T_{EM} = \frac{3\,I_R'^2\,R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Photovoltaic Systems

$$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \qquad I_{AK} = I_0\left(e^{\frac{V_{PV}+I_{PV}R_S}{K_I v_T}} - 1\right)$$

Power Flow in Lines and Cables

Cable Parameters

$R'_{LF} = \frac{1}{\sigma_C \pi r_C^2} + \frac{1}{\sigma_C 2\pi r_O t_O}$ $L' = \frac{\mu}{2\pi} ln\left({}^{r_O}/_{r_C}\right)$ $C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{r_O}/_{r_C}\right)}$ $G' = \frac{2\pi\sigma_I}{ln\left({}^{r_O}/_{r_C}\right)}$

OHL Parameters (approximate form)

$R'_{LF} = \frac{2}{\sigma_C \pi r_C^2}$ $L' = \frac{\mu}{\pi} ln\left({}^{d}/_{r_C}\right)$ $C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{d}/_{r_C}\right)}$ $G' = \frac{\pi\sigma_I}{ln\left({}^{d}/_{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{V_S^2}{Z_{SR}} cos(\theta) - \frac{V_R V_S}{Z_{SR}} cos(\theta + \delta) \qquad Q_S = \frac{V_S^2}{Z_{SR}} sin(\theta) - \frac{V_R V_S}{Z_{SR}} sin(\theta + \delta)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$

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1. This question covers several topics and all parts should be attempted.

a) Consider a national-scale AC electricity system.

i) For an example country, the UK for instance, outline the changes in proportions of generation from various sources in the last decade and the reasons for the change. [4]

ii) Explain why AC is used for almost all parts of an electricity system. [4]

b) Consider a typical photovoltaic panel.

i) Explain the factors that mean that a photovoltaic panel does not produce its rated output continuously. [4]

ii) Sketch the shape of a current against voltage curve for a photovoltaic panel for two different temperatures and explain why the temperature dependency occurs. [4]

c) Consider Buck and Boost SMPS.

i) State the assumptions that underlie the derivation of the expression for output voltage in continuous conduction mode, i.e., $\frac{V_O}{V_I} = \delta$. [4]

ii) Draw the shape of the current that flows in the inductor, diode, and capacitor of the Boost SMPS for continuous operation. [4]

iii) Explain why, for a similar selection of inductor, capacitor and switching frequency, a boost SMPS would be expected to have a higher output voltage ripple than a buck SMPS. You may consider the case of continuous conduction. [4]

d) Consider overhead lines and cables

i) A single-phase overhead line has an impedance of $1 + j1.5\ \Omega$ and a sending-end voltage of 11 kV. It supplies a load of $18 + j1\ \Omega$. Calculate the voltage at the load and the real and reactive power it consumes. [4]

ii) The same line and sending-end voltage are used to supply a constant power load of 8 MW and 1 MVAr. Estimate the difference in voltage magnitude between the sending- and receiving-end voltages. [4]

iii) Sketch the shape of the graph of power against load angle (angle difference between sending- and receiving-end voltages) for an overhead line that is inductance dominated and use this to explain the angle limit on power that a line can transmit. [4]

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2.

A **boost** switch-mode power supply (SMPS) is to be used to convert 25 V to 100 V. The SMPS is intended to operate at a maximum output current of 4 A.

The MOSFET to be used has an on-state resistance of $R_{DS(on)} = 50$ mΩ and turn-on and turn-off energy losses of $E_{Sw\text{-}on} = 55$ μJ and $E_{Sw\text{-}off} = 35$ μJ.

The diode to be used has an on-state voltage of $V_{AK(on)} = 0.95$ V.

The MOSFET is attached to a heatsink with a thermal resistance (sink to air) of 2.5 K/W, the MOSFET itself has a thermal resistance (junction to case) of 0.4 K/W and the mounting washer has a thermal resistance (case to sink) of 0.1 K/W. An ambient temperature of 40°C will be considered and the junction temperature should not exceed 100°C

You may assume that the SMPS operates in continuous conduction mode for the questions that follow.

a)

i) Calculate the duty-cycle with which the SMPS should operate. [2]

iii) Calculate the average value of the current in the inductor when the SMPS is operating at its maximum output current. [2]

Should be (ii)

b)

i) Calculate the maximum switching frequency with which the MOSFET can be operated without exceeding its maximum permissible junction temperature.[8]

ii) Calculate a value for inductor such that the ripple current in the inductor will be 0.5 A when operated at the frequency found in (i). [4]

iii) Assume that the values found so far are used and choose a capacitor such that the voltage ripple components across its capacitance and effective series resistance are 20 mV and 100 mV respectively when operating at full output current. [4]

c)

i) The control circuit of the SMPS is known to consume 1.25 W. Calculate the power losses in the MOSFET and diode (you may ignore losses in the resistance of the inductor and capacitor) for operation at maximum output power and hence calculate the efficiency of the SMPS at this point. [4]

ii) Sketch a graph of how efficiency is expected to change with output power and include how each of the power loss terms scale with output power (you should assume that the switching energy losses, E_{on}, E_{off} , scale linearly with current). [3]

(iii) Treat the power loss terms as a quadratic with coefficients a, b and c, and find the output power at which efficiency is maximum in terms of these coefficients and hence calculate this power. [3]

3. This question addresses several aspects of induction machines.

a)

i) The rotor of an induction machine has no electrical connection to an external supply and no permanent magnets. Explain why this is a desirable feature and also explain how it is that torque is exerted on the rotor. [6]

ii) What are the causes of inefficiency in an induction machine and how are they represented in the equivalent circuit. [4]

b) A three-phase induction machine with 3 pole-pairs has the following equivalent circuit parameters:

Stator resistance, $R_S = 8.0\ \Omega$.

Stator leakage reactance, $X_S = 20\ \Omega$;

Magnetising resistance, $R_M = 5000\ \Omega$;

Magnetising reactance, $X_M = 500\ \Omega$;

Referred rotor leakage reactance, $X_R = 20\ \Omega$;

Referred rotor resistance, $R_R = 6.4\ \Omega$;

The machine is supplied at a phase voltage of 3,300 V and a frequency of 50 Hz.

When used to drive a particular mechanical load, the machine runs at 960 rpm.

i) Calculate the slip. [2]

ii) Give an expression for the total input impedance of the machine and calculate its value in complex form. [6]

iii) Calculate the electro-magnetic torque. [8]

iv) Calculate the efficiency. [4]

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